

2016-07-07 TRAFFIC ANALYSIS EXERCISE - ANSWERS

Information on the infected host:

- Date/Time of the infection: 2016-07-06 00:12 UTC
- IP address of the infected Windows host: 172.16.1.126
- MAC address of the infected Windows host: 00:50:8b:ab:2e:3c (HewlettP_ab:2e:3c)
- Host name of the infected Windows host: Chevotarevich-PC
- User of the infected Windows host: Nick Chevotarevich

Malicious email that infected the Windows computer:

- Date/time: 2016-07-05 13:52 UTC
- From: FedEx 2Day A.M. <vincent.hawkins@ukmarqueewarehouse.co.uk>
- To: nick.chevotarevich@saigonroulettesolutions.com
- Subject: NICK, Problems with item delivery, n.00000611231
- Attachment: FedEx_ID_00000611231.zip
- Extracted file: FedEx_ID_00000611231.doc.js
- Analysis of the extracted .js file: [link](#)

Details:

Date/Time	Subject	Sender (r	Attachment or link
2016-07-05 09:43 UTC	Rechnun	mpsmobi	44001_Rechnung_2016-947796_20160705.zip
2016-07-05 11:55 UTC	Re: Urgel	OUTLOC	No attachment or link
2016-07-05 12:10 UTC	Scanned	<desk925	05-07-2016_rndnum(4,9)}}.docm
2016-07-05 13:52 UTC	NICK, Prc	FedEx 2D	FedEx_ID_00000611231.zip
2016-07-05 14:46 UTC	HM Reve	"web@su	http://www.bodykitskingdom.co.uk/magento/js/log.php
2016-07-05 17:37 UTC	Nota Fisc	nick.chev	https://bitly.com/29hDz6e

Shown above: The email attachment that infected the computer, highlighted in orange.

For the IP and host name, keep in mind that NBNS has a 15-character limit, so the NBNS hostname cuts off before the "C" in CHEVOTAREVICH-PC.

Time	Src	port	Dst	port	Info
2016-07-06 00:10:39	172.16.1.1	67	172.16.1.126	68	DHCP ACK - Transaction ID 0x919caaa3
2016-07-06 00:10:39	172.16.1.126	137	172.16.1.255	137	Registration NB CHEVOTAREVICH-P<00>
2016-07-06 00:10:39	172.16.1.126	137	172.16.1.255	137	Registration NB WORKGROUP<00>
2016-07-06 00:10:39	172.16.1.126	137	172.16.1.255	137	Registration NB CHEVOTAREVICH-P<20>
2016-07-06 00:10:39	172.16.1.126	49334	224.0.0.252	5355	Standard query 0x9f7b ANY Chevotarevich-PC
2016-07-06 00:10:39	172.16.1.126	137	172.16.1.255	137	Name query NB ISATAP<00>
2016-07-06 00:10:39	172.16.1.126	49334	224.0.0.252	5355	Standard query 0x9f7b ANY Chevotarevich-PC
2016-07-06 00:10:40	172.16.1.126	137	172.16.1.255	137	Registration NB CHEVOTAREVICH-P<20>
2016-07-06 00:10:40	172.16.1.126	137	172.16.1.255	137	Registration NB WORKGROUP<00>
2016-07-06 00:10:40	172.16.1.126	137	172.16.1.255	137	Registration NB CHEVOTAREVICH-P<00>
2016-07-06 00:10:40	172.16.1.126	137	172.16.1.255	137	Name query NB ISATAP<00>
2016-07-06 00:10:40	172.16.1.126	52820	224.0.0.252	5355	Standard query 0xbcdf ANY Chevotarevich-PC
2016-07-06 00:10:40	172.16.1.126	52820	224.0.0.252	5355	Standard query 0xbcdf ANY Chevotarevich-PC
2016-07-06 00:10:40	172.16.1.126	60666	172.16.1.1	53	Standard query 0x13d3 A torado.ipv6.microsoft

Shown above: The IP address and host name of the infected Windows host.

2016-07-07 TRAFFIC ANALYSIS EXERCISE - ANSWERS

Filter:	udp.port eq 67	Expression...	Clear	Apply	Save
Time	Src	port	Dst	port	Info
2016-07-06 00:10:39	172.16.1.1	67	172.16.1.126	68	DHCP ACK - Transaction ID 0
2016-07-06 00:10:42	172.16.1.126	68	255.255.255.255	67	DHCP Inform - Transaction ID 0
2016-07-06 00:10:42	172.16.1.1	67	172.16.1.126	68	DHCP ACK - Transaction ID 0
2016-07-06 00:10:45	172.16.1.126	68	255.255.255.255	67	DHCP Inform - Transaction ID 0
Seconds elapsed: 0 Bootp flags: 0x0000 (Unicast) Client IP address: 172.16.1.126 (172.16.1.126) Your (client) IP address: 0.0.0.0 (0.0.0.0) Next server IP address: 0.0.0.0 (0.0.0.0) Relay agent IP address: 0.0.0.0 (0.0.0.0) Client MAC address: HewlettP_ab:2e:3c (00:50:8b:ab:2e:3c) Client hardware address padding: 00000000000000000000 Server host name not given					

Shown above: MAC address of the infected Windows host, per DHCP traffic.

Filter:	http.request	▼	Expression...	Clear	Apply	Save
Time	Dst	port	Host	Info		
2016-07-06 00:10:45	23.218.156.26	80	www.msftncsl.com	GET /ncsl.txt HTTP/1.1		
2016-07-06 00:12:05	184.107.174.122	80	bapanivato.abjibapanichhatedi.org	GET /counter/?ad=12DgfDY8pU2cp		
2016-07-06 00:12:06	184.107.174.122	80	bapanivato.abjibapanichhatedi.org	GET /counter/?ad=12DgfDY8pU2cp		
2016-07-06 00:12:07	184.107.174.122	80	bapanivato.abjibapanichhatedi.org	GET /counter/?ad=12DgfDY8pU2cp		
2016-07-06 00:12:07	184.107.174.122	80	bapanivato.abjibapanichhatedi.org	GET /counter/?ad=12DgfDY8pU2cp		
2016-07-06 00:12:10	184.107.174.122	80	bapanivato.abjibapanichhatedi.org	GET /counter/?ad=12DgfDY8pU2cp		
2016-07-06 00:12:12	185.118.67.195	80	185.118.67.195	GET /28KNCpA1OQ5%2b7R6x%2b		
2016-07-06 00:12:21	185.118.67.195	80	185.118.67.195	GET /favicon.ico HTTP/1.1		
2016-07-06 00:13:08	90.149.107.130	80	90.149.107.130	POST / HTTP/1.1 (application/x-ww		
2016-07-06 00:13:36	75.126.29.96	80	tags.tagcade.com	GET /2.0/12025/adtag.js HTTP/1.1		
2016-07-06 00:13:39	75.126.29.96	80	serve.tagcade.com	GET /ads/tags/slot/24878?ver=2&r		
2016-07-06 00:13:39	75.126.29.96	80	serve.tagcade.com	GET /ads/tags/slot/24876?ver=2&r		
2016-07-06 00:13:40	75.126.29.96	80	serve.tagcade.com	GET /ads/tags/slot/24877?ver=2&r		
2016-07-06 00:13:41	151.101.44.166	80	tag.contextweb.com	GET /TagPublish/getjs.aspx?action=		
2016-07-06 00:13:42	151.101.44.166	80	tag.contextweb.com	GET /TagPublish/getjs.aspx?action=		
2016-07-06 00:13:42	151.101.44.166	80	tag.contextweb.com	GET /TagPublish/getjs.aspx?action=		
2016-07-06 00:13:42	151.101.44.166	80	tag.contextweb.com	GET /TagPublish/getjs.aspx?action=		
2016-07-06 00:13:42	185.86.139.19	80	ww1551.smartadserver.com	GET /ac?out=js&nwid=1551&siteid=		
2016-07-06 00:13:42	75.126.29.96	80	log.tagcade.com	GET /log?type=display&event=som		

Shown above: The exercise pcap, filtered in Wireshark.

CNT	Src IP	SPort	Dst IP	DPort	Event Message
3	172.16.1.126	49158	184.107.174.122	80	ET TROJAN JS/Nemucod requesting EXE payload 2016-03-31
8	184.107.174.122	80	172.16.1.126	49158	ET POLICY PE EXE or DLL Windows file download
5	184.107.174.122	80	172.16.1.126	49158	ET MALWARE Windows executable sent when remote host claims...
7	184.107.174.122	80	172.16.1.126	49158	ET POLICY Binary Download Smaller than 1 MB Likely Hostile
4	184.107.174.122	80	172.16.1.126	49158	ET TROJAN JS/Nemucod.M.gen downloading EXE payload
4	184.107.174.122	80	172.16.1.126	49158	ET CURRENT_EVENTS Likely Evil EXE download from MSXMLHTTP ...
1	184.107.174.122	80	172.16.1.126	49158	GPL WEB_CLIENT Internet Explorer URLMON.DLL Content-Encod...
1	172.16.1.126	49160	185.118.67.195	80	ET TROJAN Miuref/Boaxxe Checkin
2	172.16.1.126	49165	173.83.171.211	443	ET TROJAN Win32.Kovter Client CnC Traffic

2016-07-07 TRAFFIC ANALYSIS EXERCISE - ANSWERS

Shown above: EmergingThreats alerts from the pcap.

```
[**] [1:38963:1] MALWARE-OTHER Win.Trojan.Nemucod file download [**]  
07/06-00:12:05.768108 172.16.1.126:49158 -> 184.107.174.122:80  
  
[**] [1:15306:22] FILE-EXECUTABLE Portable Executable binary file magic detected [**]  
07/06-00:12:05.848547 184.107.174.122:80 -> 172.16.1.126:49158  
  
[**] [1:25061:7] FILE-EXECUTABLE Microsoft Software Installer MSI binary file magic detected [**]  
07/06-00:12:05.848547 184.107.174.122:80 -> 172.16.1.126:49158  
  
[**] [1:38963:1] MALWARE-OTHER Win.Trojan.Nemucod file download [**]  
07/06-00:12:06.695997 172.16.1.126:49158 -> 184.107.174.122:80  
  
[**] [1:648:18] INDICATOR-SHELLCODE x86 N00P [**]  
07/06-00:12:07.148259 184.107.174.122:80 -> 172.16.1.126:49158  
  
[**] [1:23861:11] FILE-OTHER heapspray characters detected - binary [**]  
07/06-00:12:07.459143 184.107.174.122:80 -> 172.16.1.126:49158  
  
[**] [119:15:1] (http_inspect) OVERSIZE REQUEST-URI DIRECTORY [**]  
07/06-00:12:12.438560 172.16.1.126:49160 -> 185.118.67.195:80
```

Shown above: Some of the Snort alerts from the pcap using the subscriber ruleset.